

NASSCOM VIRTUAL INTERNSHIP PROGRAM

PROJECT REPORT

{This is a comprehensive report on what is learnt since the beginning of the internship.}

UNIVERSITY MAHARANIS COLLEGE

Team members

MEMBER	ROLL NO.
ISHA SAINI	
BALJEET KAUR	
KANAK	

Name of the course :- AI SPECIALIST IN GROWTH MARKETING

Project assigned :- AI POWERED VIDEO SCRIPT GENERATOR

Mentor's Name _____

Mentor's Signature _____

Submission date :- _____

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to NASSCOM for providing me with the opportunity to participate in the Virtual Internship Program. This internship has been a valuable learning experience that allowed me to gain practical knowledge and develop technical skills by working on a real-world project.

I would also like to thank my mentors and trainers for their continuous guidance, support, and valuable suggestions throughout the internship. Their encouragement and feedback helped me understand the project development process and successfully complete my work.

I extend my heartfelt thanks to my college faculty and project guide for their constant motivation, guidance, and support during the internship and project preparation.

Finally, I would like to thank my family and friends for their encouragement, patience, and moral support throughout this journey. Their support motivated me to complete the internship successfully.

This internship has enhanced my understanding of Artificial Intelligence, project development, teamwork, and problem-solving, and I am grateful for the opportunity to learn and grow.

WHAT IS ARTIFICIAL INTELLIGENCE ?

When the computers perform the tasks that require human intelligence , it's called artificial intelligence. for example **speech recognition** , **face recognition in our mobile phones** , **predictions** , **the YouTube algorithm and recommendation** is also an example of artificial intelligence only. It is a discipline in computer science.

How does artificial intelligence works ?

AI works by learning patterns from data and then using those patterns to make predictions or decisions when it receives new input.

WHAT IS MACHINE LEARNING AND AI MODEL ?

Machine learning is further a subset of artificial intelligence. **Machine Learning (ML)** is a method of **creating AI**. Teaching computers to learn patterns from data instead of writing every rule manually. Machine Learning is the process used to train a mathematical model on data. After training, that trained model becomes an AI model that can make predictions.

Imagine you want a computer to identify whether an email is spam or not . then in :-

Traditional Programming

A programmer writes rules:

IF email contains "WIN MONEY"
→ Spam

IF email contains "FREE PRIZE"
→ Spam

IF email contains "CLICK HERE NOW"
→ Spam

Now in ML , Instead of writing rules:

we collect data:

EMAIL	SPAM?
WIN MONEY NOW	Yes
FREE PRIZE	Yes
MEETING AT 5 PM	No
PROJECT REPORT	No

This is called **training data**.

The programmer still writes code. For example:

```
Create a model.  
Read training data.  
Find patterns.  
Adjust parameters.
```

This code creates a mathematical structure.

Initially:

Model = Random

It knows nothing.
The model contains numbers.
Learning = changing these numbers.

The model has learned patterns.

```
Money = +2.5  
Prize = +2.1  
Meeting = -1.8  
Project = -2.0  
...
```

This trained mathematical structure is the **AI model**.

What Does a Mathematical Model Actually Look Like?

Not like a robot.

Not like a human brain.

Internally it is mostly a giant collection of numbers. For example:

0.72 , -1.51 , 3.84 , 0.01 , 7.22

...

Millions or billions of such numbers.

Before: Random Numbers

After training : Meaningful Numbers

Now the model understands patterns, understands the relationship between different data.

This trained model becomes: AI Model

What Is An AI System?

An AI System is bigger than the model.

Example:

ChatGPT Website/App

+

Servers

+

Databases

+

Security Systems

+

AI Model

+

User Interface

All together:

AI System

The AI model is just one part.

For ChatGPT: The AI model which works behind is :-

Large Language Model (LLM)

Now :-

Chat Interface

+

Servers

+

Databases

+

Safety Systems

+

AI Model

Together:

ChatGPT System

Types of machine learning :-

1] **Supervised learning** - models learn from **labeled data**. Where both input and output data exists and they are in a structured manner . Also in mathematical terms it is ' $y = f(x)$ ' where y

is the output and predicted on the basis of x which is input , or we can say that y is the function of x.

Note :- the supervised learning can be used to solve two types of problems:-

- 1) **classification/ classes problems** where we map our input in a way to give us pre defined output. Example:- email can be divided into spam or not spam . similarly the bank loan for an applicant can either be approved or not approved . **simply we have finite no. of classes available for us** . The classification can further be divided into binary and multi class classification like **sentiment analysis** .

Classification means predicting a category or class.

The output is **not a number**, but a label.

Examples:

- Spam or Not Spam
- Cat or Dog
- Fraudulent Transaction or Genuine Transaction
- Disease Present or Not Present

Notice that the answer always belongs to one of the predefined categories.

- 2) **Regression problems** means predicting a numerical value.

Instead of choosing a category, the model predicts a number.

Examples:

- Predict the price of a house → ₹52,00,000
- Predict tomorrow's temperature → 32°C
- Predict a student's exam score → 88 marks

Here, the output is always a **number**.

2] **Unsupervised learning** - **models learn from raw data** , where patterns are recognised . Here, the model is given only the input data. There are **no correct answers (labels)**.The model tries to discover hidden patterns or relationships by itself. example :-

Suppose an online shopping company has customer purchase data but doesn't know which customers are similar. The model automatically groups customers based on their shopping habits.

*There are certain data points different from other data referred to as **outliers or anomaly** .
useful in the field of medical , cybersecurity , finance .

Note : - problems such as **clustering and association** can be solved with the help of unsupervised learning .

1) Clustering :- **Clustering means grouping similar data together.**

The model creates groups based on similarity.

Example:

A shopping website groups customers into:

- Students
- Working professionals
- Senior citizens

No one tells the model these groups—it finds them on its own.

2) Association:- **Association means finding items that are frequently related to each other.**

The model discovers patterns like:

"People who buy X also often buy Y."

Examples:

- People who buy a laptop often buy a laptop bag.
- People who buy bread often buy butter.
- People who buy a mobile phone often buy a phone case.

This is why shopping websites recommend related products.

WHAT IS DEEP LEARNING ?

Till now the ML algorithms about which we have talked about :- is statistical ML{ we had data in the tabular form } . but **when we have unstructured data** { such as youtube video , human faces then the deep learning performs better }. Deep Learning is a **subset of Machine Learning** that is specially designed to work with **large amounts of unstructured data**, such as:

- Images ,Videos ,Audio ,Text , Human speech

In traditional machine learning, we usually work with **structured (tabular) data**, where the information is organized into rows and columns.

But , when giving a computer a **human face** or a **YouTube video**, we should understand :-

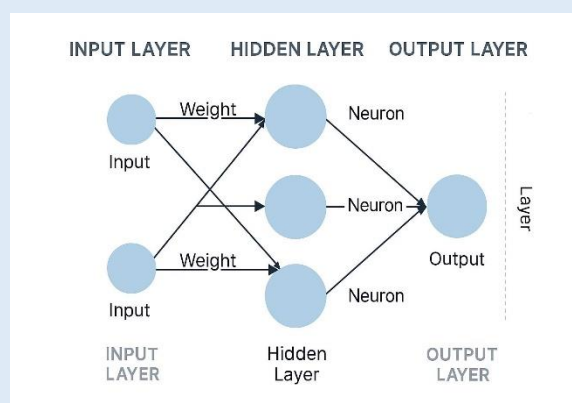
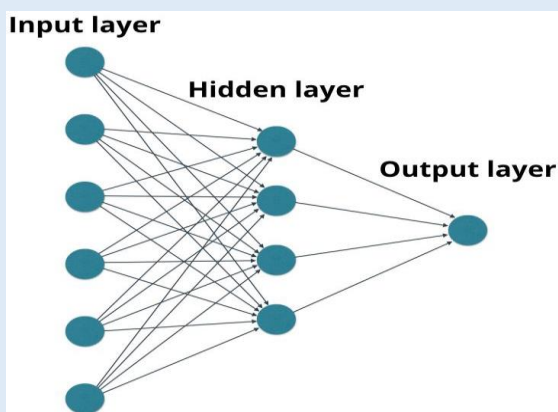
A face is made up of **millions of pixels**, and a video contains **thousands of images (frames)**. This data is too complex for many traditional machine learning algorithms to handle effectively.

This is where **Deep Learning** performs much better.

when we input our data into the model it passes through input layer of neural network , then there are many hidden layers and then the output layer . the neurons are interconnected with each other.

What is a Neural Network?

A Neural Network is a mathematical model that is **inspired by the human brain**. Just as our brain contains billions of interconnected neurons, a neural network contains many **artificial neurons** connected together. These neurons work together to learn patterns from data.



WHAT IS GENERATIVE AI

Generative AI can be defined as **AI that can produce new content based on the data it is trained on** .

Example :- **chat gpt (open ai chatbot application) can produce text data.**

Similarly we have applications producing image, video , audio and other synthetic data .

OTHER EXAMPLES ARE AS FOLLOWS :-

AI Model	User-Facing Applications/Tools	Output
GPT (OpenAI)/ LLMs	ChatGPT, Microsoft Copilot	Text, code, reasoning
Gemini (Google)	Google Gemini	Text, code, reasoning
Claude (Anthropic)	Claude	Text, analysis, writing
DALL·E	ChatGPT image generation	Images
Imagen	Gemini image features, Google AI image tools	Images
Nano Banana (Google image model)	Gemini image-generation features, Google AI Studio experiments	Images, image editing

Stable Diffusion	DreamStudio, Automatic1111, ComfyUI, Fooocus, many AI image apps	Images
Midjourney Model	Midjourney	Images
Sora	OpenAI video-generation tools	Videos
Veo	Google AI video tools	Videos
ElevenLabs Voice Models	ElevenLabs apps	Speech, voice cloning
Suno Models	Suno	Songs, music
Udio Models	Udio	Songs, music

HUGGING FACE (told in class)
Hugging Face
Stores AI models
So Hugging Face is often called the
"GitHub of AI models."

Chatbot :- a smart computer program
or AI that can talk to humans through
text or voice .
Ex:- open AI Chat GPT, Microsoft
Copilot, google gemini , perplexity ,
claude , etc.

HOW IS TRADITIONAL AI DIFFERENT FROM GENERATIVE AI?

Real AI is mostly: pattern recognition , prediction , automation

Traditional AI mainly:

Analyzes data, finds patterns, makes decisions, or predictions.

It usually does **not create new content**.

The recommendation system of youtube is a classic example of **traditional AI / machine learning**.

But as the name suggests generative AI can generate new data .

MEANING OF LLMs ?

LLMs stands for large language model , it an AI model trained on large amount of textual data so it can understand and generate human data . It works with enormous datasets , books , websites, articles , with language : hindi , english , questions , conversations and so on .

Example:- Chat GPT uses LLM model .

The LLMs have many parameters .

What are parameters ?

The internal numerical values that an AI model learns during training. These values store the patterns and knowledge the model has learned from data.
So , in an AI model, the learned information is stored in parameters.
The model uses its parameters to predict.

Transformer Architecture

A Transformer **is a** deep learning architecture (or framework) that is specially designed to **understand and process** text.

Earlier AI models had difficulty handling long pieces of text efficiently.

Transformers were designed to:

- understand relationships between words
- process large amounts of text efficiently
- handle long contexts better

Transformer = The design or architecture used to build the model.

Large Language Model (LLM) = The final AI model that is built using that architecture and trained on huge amounts of text.

INTRODUCTION TO MARKETING AND BASIC CONCEPTS :-

{For nasscom virtual internship program , we as a team was enrolled in the course “AI SPECIALIST IN GROWTH MARKETING”. In the sessions we attended , we were also explained to basics of marketing and use of AI in growth marketing . }

- **Marketing** is the process of **understanding customer needs, creating value, and promoting products or services to attract and retain customers.**
- Marketing concepts :- product differentiation , quality differentiation { unique selling point / usp } , price differentiation .

- **7 P's of marketing or marketing mix** { 4 for tangible products + 3 for non tangible products } :- **pillars of marketing strategy** . * product , price(revenue model / penetration model) , place , promotion + people , process , physical evidence.
- concept of SWOTs and swot analysis / metric :- can be done for a company / organization even at individual level . * strength and weakness{ internal factors } , opportunities and threats { external factors } .google sheets , excel sheets .
- A **pricing strategy** is the **approach or plan** a company uses to decide the price of its product or service in order to achieve a business goal (such as increasing sales, maximizing profit, or attracting customers).
- **Pricing models** :- **skimming pricing , penetration pricing , value based pricing , dynamic pricing model , competition based pricing , psychological pricing , premium pricing .**
- Price Elasticity of Demand **measures** how much the demand for a product changes when its price changes. **Inelastic Demand** :- the change in price has little effect on demand . Suppose the price of a life-saving medicine increases from ₹100 to ₹120. Most people will still buy it because they need it.
- **Sales V/S Marketing** :- both are business functions or processes with the goal to generate maximum revenue .

Marketing is the process of creating awareness, attracting customers, and generating interest in a product or service.

Example: Running social media ads, email campaigns, or TV advertisements to promote a new smartphone.

Sales:- Sales is the process of convincing customers to purchase the product or service. It is the subset of marketing .

Example: A salesperson explaining the features of the smartphone and completing the purchase.

Marketing	Sales
Attracts potential customers	Converts potential customers into buyers
Creates awareness and interest	Closes the deal and generates revenue
Happens before the purchase	Happens at the time of purchase

- **Product lifecycle** :- Introduction{ expenses maximum } , growth , maturity { maximum sales } , decline .
- **5 Cs of Marketing** :- All marketing decisions are taken on these 5Cs , **situational analysis framework** . It helps a company to understand its current situation and answer important questions before launching or promoting a product.

Company , Customers , Collaborators , Competitors , Climate(govt. rules and regulations , seasonality of product / services , social trends , emerging technologies in market) . { different kind of questions are asked at each level }

- A **Lead Score** is a numerical value assigned to a potential customer (lead) based on how likely they are to become a paying customer.

The higher the score, the more likely the lead is to make a purchase.

Example:

- Rahul visited the website, downloaded a brochure, and requested a demo → Lead Score = 90/100 (High chance of buying)
- Priya only visited the homepage once → Lead Score = 20/100 (Low chance of buying)
- A **Lead Scoring Model** is the **set of rules or criteria used to calculate a lead's score**.
- It decides **how many points** should be given for different customer actions or characteristics.

* used by sales team:- for conversion of prospect to customer . lead can be stated as potential customer . * The sales team can provide customized offers to different sections of customers .

- **STP Framework in marketing** :- The **STP Framework** is a marketing strategy used by businesses to **identify the right customers and market their products effectively**.

STP stands for Segmentation , Targeting , Positioning

Segmentation: **Segmentation** means **dividing the market into smaller groups** based on similar characteristics.

- **Demographic** → Age, gender, income, education.
- **Geographic** → Location.
- **Online Presence** → Digital behavior and online activity.
- **Psychographic** → Lifestyle, values, interests.
- **Behavioral** → Buying habits and loyalty.

Example :- A clothing brand creates separate products for children, teenagers, and adults.

Targeting: Targeting means selecting the specific customer segment that the company wants to focus on.

- **Measurability** → Can we measure the market?
- **Sustainability** → Is it profitable in the long run?
- **Accessibility** → Can we reach the customers?
- **Actionability** → Can we effectively serve them?

Example :- A luxury watch brand targets high-income customers instead of everyone

Positioning: Positioning means creating a unique image of the product in the customer's mind so that it stands out from competitors.

Example :- Volvo positions itself as a safe and reliable car brand.

- **Consumer adoption process framework** :- awareness, interest , evaluation , product trial , adoption .
- **Ansoff matrix / product market expansion grid:-** The Ansoff Matrix is a strategic planning tool that helps businesses decide how to grow their business using products and markets. Market penetration, product development , market development ,diversification.

Market Penetration

Sell existing products to existing customers.

The goal is to **increase market share.**

Example: Coca-Cola offers discounts and advertisements to encourage more people to buy Coca-Cola.

Product Development

Develop new products for existing customers.

Example: Apple launches a new iPhone model for its current customers.

Market Development

Sell existing products in new markets.

This could mean entering:

- A new city
- A new country
- A new customer segment

Example: An Indian food brand starts selling its products in Europe.

Diversification

Introduce new products in new markets.

This is the **riskiest strategy** because both the product and the market are new.

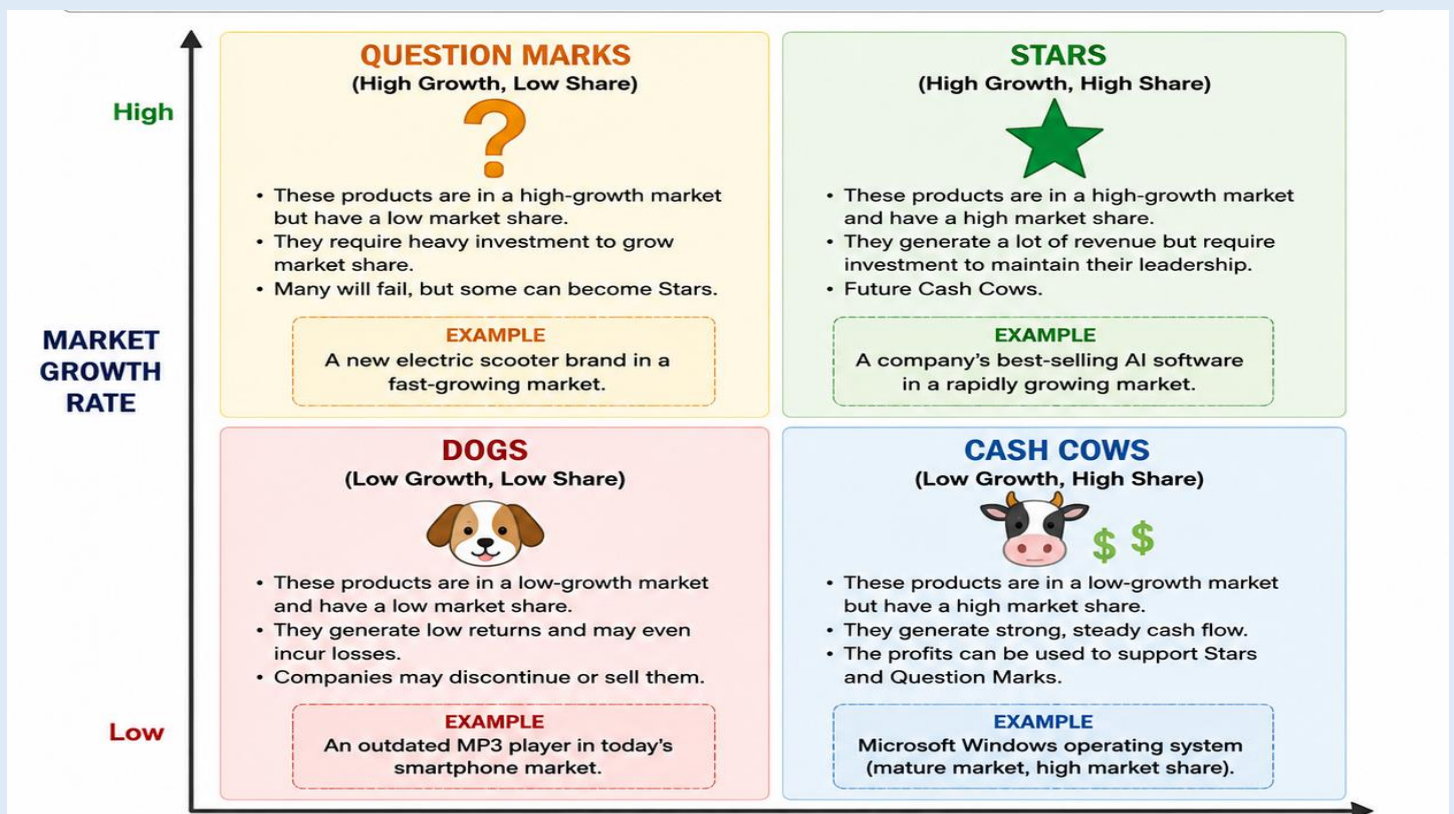
Example: Tesla, known for electric cars, expanding into home energy storage systems.

- **BCG metrix** :-helps companies to decide how to prioritize their different businesses
The BCG Matrix is a business analysis tool developed by the Boston Consulting Group (BCG).

It helps companies prioritize their different products or business units based on:

- Market Growth → How fast the market is growing.
- Relative Market Share → How strong the company's position is compared to competitors.

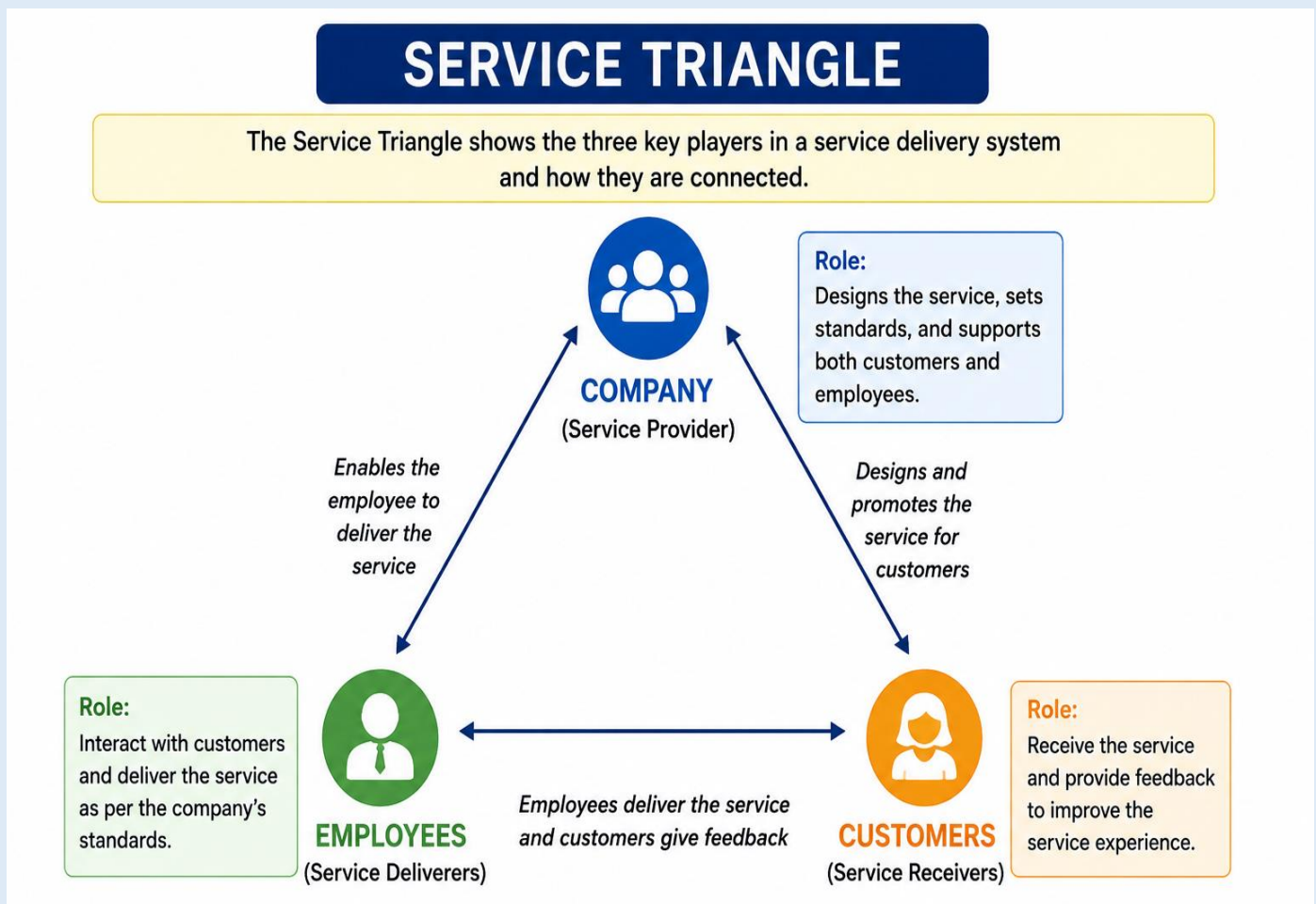
The BCG Matrix has four categories.



- **Ambush marketing** :- a strategy that involves a brand team trying to associate their products or services with a big event that already has official companies - sponsors .
- **5 As of marketing** :- framework for converting the strangers into customers .

Awareness, Appeal , Ask , Act , Advocate.

- Gap in the Market :- A **market gap** is an **unmet customer need or an opportunity that competitors are not fulfilling**. Companies identify these gaps and position their products to meet those needs. **Example:** Before food delivery apps became popular, people had limited options to order food from restaurants. Companies like Zomato and Swiggy identified this **market gap** and built services around it.
- Marketing V/S Advertising { part of marketing } :- **advertising benefit is always greater**



to a well marketed brand .

WHAT IS GROWTH MARKETING ?

Growth Marketing = Marketing + Data Analysis + Experimentation

Instead of only trying to get new customers, growth marketers also focus on:

- Getting more users
- Increasing sales
- Retaining customers

- Encouraging referrals
- Improving customer lifetime value

Example

Suppose you run an online store. A traditional marketer might: Run Instagram ads

A growth marketer might:

- Test different ad designs , test different landing pages , send email reminders for abandoned carts, analyze which customers buy repeatedly , create referral programs .

Everything is measured and optimized using data.

AI in growth marketing

1. Creating Ad Creatives

AI can generate:

- Ad copy
- Images
- Videos
- Headlines

*Instead of manually writing 20 ad headlines, AI generates 100 variations in seconds.

2. Customer Segmentation

AI groups customers automatically.

Example:

Customer Type	Behavior
New Visitors	First visit
Frequent Buyers	Purchase often
Cart Abandoners	Leave without buying

The marketer can then send different messages to each group.

3. Predicting Customer Behavior

AI can predict:

- Who is likely to buy
- Who may stop using the product
- Which customers are valuable

Example:

An e-commerce company may identify users who are 80% likely to purchase in the next week and target them with special offers.

4. Email Marketing Automation

AI can:

- Write emails
- Decide when to send them
- Personalize subject lines

Example:

Customer A: Still thinking about those running shoes?

Customer B: Your favorite sports products are back in stock.

Chatbots and Customer Support

5. AI chatbots

- Answer questions
- Recommend products
- Collect leads
- Help users complete purchases

6. Finding Growth Opportunities Through Data

AI can analyze:

- Website traffic
- User journeys
- Conversion rates

and tell marketers things like: "Mobile users abandon checkout 30% more often than desktop users."

A human might take hours to discover this; AI can surface it quickly.

ETHICAL AI — MEANING

Ethical AI means designing, developing, and using Artificial Intelligence in a way that is **fair, safe, transparent, responsible, and respectful of people's rights**.

In simple words:

Ethical AI = AI that does the right thing and avoids causing harm.

Key Principles of Ethical AI

1. Fairness AI should not unfairly discriminate against people.

Example:

If an AI system helps hire employees, it should evaluate candidates based on qualifications,

not unfairly favor or disadvantage people because of gender, race, or other unrelated characteristics.

2. Transparency People should have some understanding of how AI reaches important decisions.

Example:

If a bank uses AI to reject a loan application, there should be a way to explain the main reasons for the decision.

3. Privacy AI should protect users' personal information.

Example:

A healthcare AI should not share patient records without proper authorization.

4. Accountability Someone must be responsible for the AI's actions and outcomes.

Example:

If an AI system makes a harmful mistake, the organization that deployed it should investigate and address the issue.

5. Safety and Reliability AI should work as intended and avoid causing harm.

Example:

An AI-powered self-driving system should be thoroughly tested before being used on public roads.

***Fair Treatment of Consumers**

AI should not unfairly favor or disadvantage certain groups of customers.

Marketing Example

A company uses AI to decide who receives a discount coupon.

Without Ethical AI:

- Men get 20% discount offers.
- Women get only 5% discount offers.
- The company never intended this, but the AI learned a biased pattern from historical data.

This would be unfair. **How Ethical AI helps?**

The marketing team audits the AI and checks:

- Are discounts distributed fairly?
- Is any customer group being excluded without a valid reason?

Real Tools Companies using platforms such as:

- **Salesforce Einstein**
- **Adobe Sensei**

often review AI-driven audience segmentation to ensure campaigns are not unintentionally biased.

* **Responsible Use of Data** AI should only use customer data appropriately and with proper permission. Marketing Example:- Suppose an online store collects:

- Name , Email , Purchase history

Customers agreed to receive shopping recommendations. Ethical AI can use this information to recommend products. However, it should not:

- Sell personal data to unknown parties.
- Use data for purposes the customer did not agree to.

Real Example

A recommendation system like those used by:

- **Amazon**
- **Netflix**

analyzes behavior to personalize recommendations while operating under privacy policies and legal requirements.

* **Trustworthy Marketing Practices**

AI should not deceive customers. **Marketing Example :-** An AI tool generates ad copy.

Unethical AI might generate:

"This supplement guarantees weight loss in 7 days." even if there is no evidence.

Ethical AI practices require human review so marketing claims remain accurate.

Real Tools :- Marketers often use: ChatGPT , Jasper to create marketing content.

Ethical use means:

- **Fact-checking outputs.**
- **Avoiding misleading claims.**
- **Clearly distinguishing facts from marketing language.**

* Compliance with Regulations

AI-powered marketing must follow laws and regulations.

Example :- An AI email system wants to send promotions.

Before sending, the company must ensure:

- Customers consented to receive emails.
- There is an unsubscribe option.
- Personal data is handled legally.

Email marketing platforms such as:

- Mailchimp
- HubSpot

include features that help marketers manage consent, subscriptions, and compliance requirements.

SEARCH ENGINE OPTIMIZATION {SEO }

SEO is the process of improving a website so that it appears higher in search engine results (like Google) when people search for relevant topics.

Example :- Suppose you have a website that sells running shoes.

A person searches: "best running shoes for beginners" If the website appears on **Page 1 of Google**, more people will visit it. **SEO helps increase the chances of your website appearing higher in those results.**

How does seo work ?

Search engines like **Google** scan millions of websites and decide which ones are most relevant to a user's search. When someone searches: "**Best laptops under ₹50,000**"

Google looks for web pages that:

- Match the search keywords.
- Have useful and high-quality content.
- Load quickly.
- Are easy to use.
- Are trusted by other websites.

If our website meets these factors, it is more likely to appear higher in the search results.

How Is AI Used in SEO?

AI helps marketers improve their websites so they have a better chance of ranking higher.

1. Keyword Research :-AI suggests the words and phrases people are searching for.
2. Content Creation:- AI tools such as **ChatGPT** can help generate:Blog articles, Product descriptions , Website FAQs.
3. Content Optimization : AI analyzes your content and suggests improvements, such as: Adding missing keywords, Improving headings, Making the content easier to read.

AI POWERED VIDEO SCRIPT GENERATOR

{The actual project }

For the completion of any project these are the steps to be followed :- { explained by mentor in the class }

Already the individual , important pdfs have been prepared and uploaded by our team on github repository .

1. Pre – requisites , set up , configurations
2. Define problem statement , ideation , and idea prioritization
3. Requirement analysis , deep solution and technical architecture .
4. Project design milestone
5. Project planning , milestone , standup calls
6. Project development { where coding plays role }
7. Functional and performance testing
8. Mentor review and approval .

The other important things are as follows :-

The project can be broken down into these components:

<i>Component</i>	<i>Example</i>
<i>Problem</i>	Script writing takes time
<i>Users</i>	Content creators, marketers
<i>Input</i>	Topic, audience, platform, tone, duration
<i>AI Model</i>	GPT, Gemini, Claude
<i>Prompt</i>	Detailed instructions to generate a script
<i>Output</i>	Hook + Body + CTA
<i>Interface</i>	Simple form with input fields
<i>API</i>	Sends prompt to the AI model

<i>Result</i>	Displays the generated script
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Overview

"Our application takes information from the user, sends it to an **AI model (LLaMA)** through **Groq's API**, gets back a generated video script, and shows it in a web application built using **Flask**. We also use **CORS** so the frontend and backend can communicate, and we keep our secret API key safely inside a **.env file**."

Groq provides access to AI models through an API.

What is Flask?

Flask is a **Python web framework**. It helps you create a web application. Without Flask, Python is just running programs. With Flask, Python can create websites.

Example:

Browser → Flask → Python Code → Groq API → LLaMA

What is .env?

This is one of the most important concept. If Groq gives:- gsk_89asd98213jdas... This is our **API Key**. Its like a password. If someone steals it, they can use our account.

Instead of writing

```
API_KEY="gsk_xxxxxxxxx"
```

inside the code, we create

.env file Inside:-

```
GROQ_API_KEY=gsk_xxxxxxxxx
```

Now only python reads it.

- **Groq** is **not** the AI model—it provides fast infrastructure and an API to access models.
- **LLaMA** is the AI model, created by **Meta**.
- **Flask** is the backend web framework that connects your application to the AI.
- **CORS** is a browser security mechanism that allows your frontend and backend to communicate.
- **.env** is a secure configuration file for storing secrets like API keys.

PRE-REQUISITES, SETUP, CONFIGURATION

Goal

- Prepare everyone's laptop and accounts: - Install Python , Install VS Code , Install Git , Create Groq account , Clone GitHub repository
- * Everyone has the same development environment

PHASES OF PROJECT DEVELOPMENT

Define problem statement , ideation , and idea prioritization

Problem statement :- A problem statement clearly describes the issue or challenge that a project aims to solve. It explains why the project is needed and what problem users are facing.

* Many content creators and businesses spend a lot of time writing engaging video scripts for marketing and social media. Creating hooks, script flow, scene ideas, and call-to-action manually requires creativity and effort. This project solves this problem by using Artificial Intelligence to generate complete promotional video scripts quickly, helping users save time while improving content quality.

Ideation :- Ideation is the process of generating and discussing different ideas to solve a problem before selecting the best one.

*During the ideation phase, different AI-based content generation ideas were explored, such as AI caption generators, AI blog writers, AI social media post creators, and AI video script generators. After comparing these ideas, an AI-powered video script generator was selected because video content is widely used in digital marketing and requires significant effort to create manually.

Idea prioritization :- Idea prioritization means selecting the best idea based on factors such as usefulness, feasibility, impact, and ease of implementation.

*The AI-powered video script generator was prioritized because it provides practical value to marketers, businesses, and content creators. It can generate hooks, promotional scripts, scene suggestions, and call-to-action messages within seconds. The project is also technically feasible using Groq API and the Llama 3.3-70B Versatile model.

Requirement analysis , deep solution and technical architecture .

Requirement analysis is the process of identifying what is needed to build the project successfully, including functional and technical requirements.

Functional Requirements

- User enters product information and campaign details.
- AI generates a hook.
- AI generates a promotional video script.
- AI suggests visual scenes.
- AI generates a Call-to-Action (CTA).

Technical Requirements

- Python
- Flask Framework
- Groq API
- Llama 3.3-70B Versatile Model
- HTML/CSS
- VS Code
- GitHub
- python-dotenv for API key management

A **deep solution** explains how the proposed system solves the identified problem using appropriate technologies and processes.

- The project uses the Groq API with the Llama 3.3-70B Versatile language model. Users provide product details through a web interface. The Flask backend processes the input, sends prompts to the AI model, and receives generated content. The application then displays the generated hook, script, scene suggestions, and CTA to the user in a structured format.

Technical architecture describes how different components of the system work together.

Project planning , milestone , standup calls

Project planning is the process of organizing tasks, assigning responsibilities, and scheduling work to complete the project efficiently.

The project was divided into different phases, including research, requirement analysis, UI design, backend development, AI integration, testing, and deployment. Each team member was assigned specific tasks, and the work was completed according to the planned schedule.

Milestones are important checkpoints that show the completion of major project activities. Example :- research completed , frontend completed , backend completed etc.

Stand-up calls are short team meetings conducted regularly to discuss progress, completed work, upcoming tasks, and any challenges.

Regular stand-up meetings were conducted among team members to review project progress, discuss completed tasks, identify issues, and plan the next activities. These meetings helped improve communication and ensure that the project stayed on schedule.

Project development

The development of the AI-Powered Video Script Generator was carried out in a structured manner by dividing the project into different modules. The objective was to build a web application that can automatically generate promotional video content using Artificial Intelligence.

The development process began with setting up the development environment. Python was selected as the programming language because of its simplicity and strong AI support. The Flask framework was used to develop the backend of the application, while HTML and CSS were used to design a simple and user-friendly interface. Visual Studio Code (VS Code) was used as the primary code editor, and GitHub was used for version control and team collaboration.

The frontend was designed to collect all the necessary information from the user. The interface allows users to enter details such as the product name, product description, target audience, campaign goal, and preferred video duration. The design was kept simple so that users could easily interact with the application without any technical knowledge.

The backend was developed using Flask. Different API routes were created to process user requests. These routes receive the user input, validate the required fields, generate prompts, communicate with the AI model, and return the generated output to the frontend. Proper input validation was also implemented to ensure that mandatory information is provided before processing the request.

One of the most important parts of the project was the integration of the **Groq API**. The application uses the **Llama 3.3-70B Versatile** large language model through the Groq API to generate AI-based content. Instead of giving a generic request to the AI model, carefully designed prompts were created for each feature. Separate prompts were written for generating:

- A catchy Hook
- A complete promotional Video Script
- Scene Suggestions
- A Call-to-Action (CTA)

This prompt-based approach helped the AI generate more accurate, relevant, and structured responses according to the user's requirements.

The application follows a modular architecture. Separate Python functions were developed for each AI generation task, such as `generate_hook()`, `generate_script()`, `generate_scenes()`, and `generate_cta()`. These functions internally call a common AI communication function that sends the prompt to the Groq API and retrieves the generated response. This modular approach improves code readability, reduces duplication, and makes future maintenance easier.

For security purposes, the Groq API key was stored in a `.env` file instead of hardcoding it into the source code. The `python-dotenv` library was used to load the API key securely, while the `.gitignore` file ensured that sensitive information was not uploaded to the GitHub repository.

The generated outputs were displayed on the web interface in a structured format. The system produces an attention-grabbing hook, a scene-wise promotional video script, visual scene suggestions, and a concise call-to-action. These outputs help users create marketing videos more efficiently without manually writing the entire script.

Throughout the development phase, GitHub was used for source code management and collaboration among team members. Changes were committed regularly, making it easier to track progress, manage different versions of the project, and work collaboratively.

Finally, after completing the implementation of all modules, the application was tested to verify that each feature functioned correctly. Different input scenarios were used to ensure that the generated content was relevant, the application responded correctly to user requests, and the overall system performed smoothly.

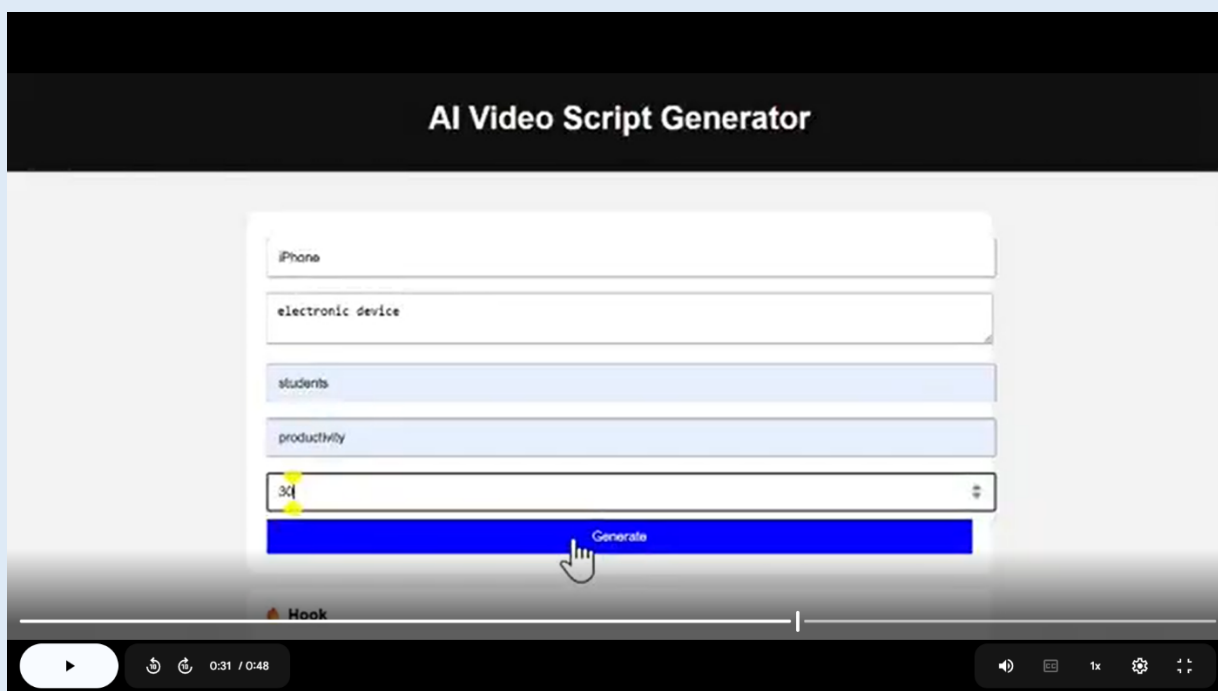
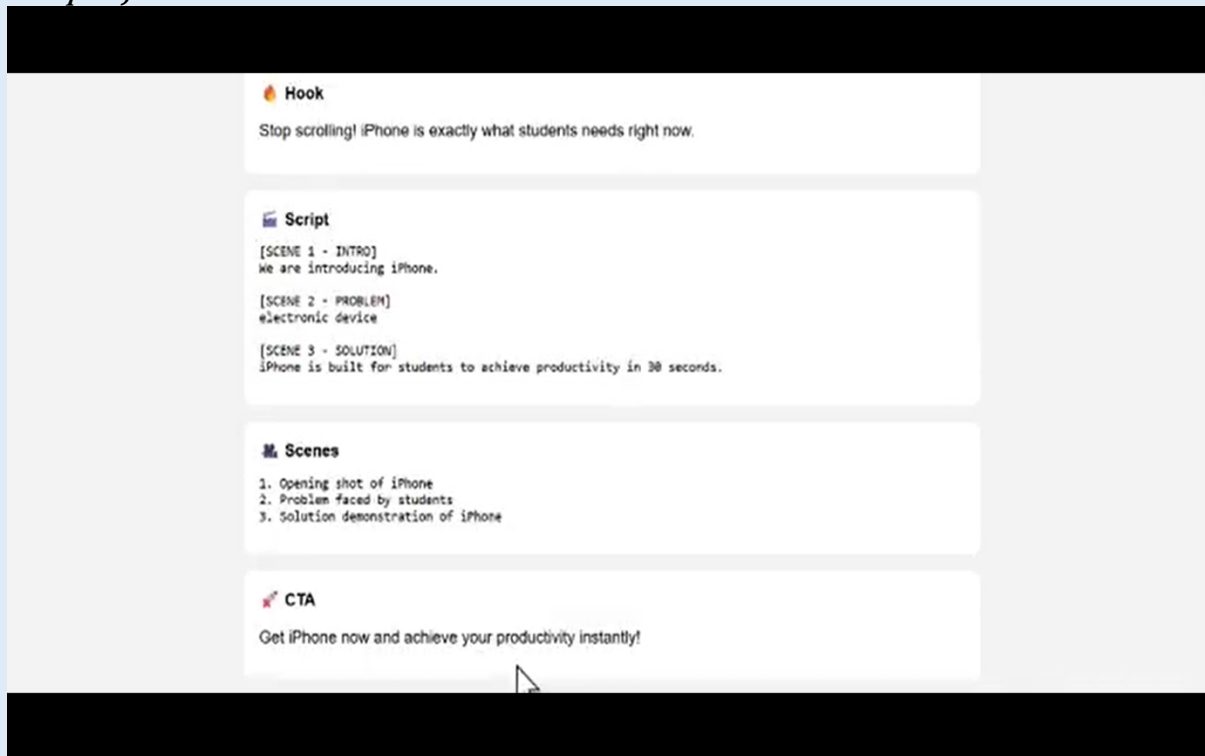
Functional and performance testing

Functional testing checks whether every feature of the application works correctly according to the project requirements.

Performance testing measures how efficiently and reliably the application performs under different conditions.

TESTIMONIALS { SCREENSHOTS }

We have attached these screenshots to let you know about our successful completion of the project.



SKILLWALLET

Dashboard

Skill Bank

Skill Wallet

Skill Card

Skill Credits

Dashboard > Skillwallet > Program > AI specialist growth marketing

0 AI Token300 Credits

Virtual Internship Program

AI Specialist Growth Marketing

40%

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100K Agentblazer Champions

6 Hrs 35 Min100%

CategoryAI Agents

LevelIntermediate

Access Resource

20 Hrs - Generative AI With Google Cloud

20 Hrs100%

CategoryGoogle Generative AI

LevelBeginner

Access Resource

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SKILLWALLET

Dashboard

Skill Bank

Skill Wallet

Skill Card

Skill Credits

myskillwallet.ai/dashboard/skillwallet/module/ai-specialist-growth-marketing-6a1026268e05517a37e491db/group-projects/6a2fbee312d2c0ba5ef35b61/VideoScript-AI-AI-Powered-Video-Content-Generator

Ask Gemini

VideoScript AI: AI-Powered Video Content Generator

OverviewWorkspaceKanban

Mentor

No mentor assigned yet

Team Members

AAnchal SharmaMember

BBaljeet KaurMember

SSakshi SainiMember

KKanak DhindhwalTeamLead

IIsha SainiMember

Project Stats

Epics

Total Epics: 8

Stories & Subtasks

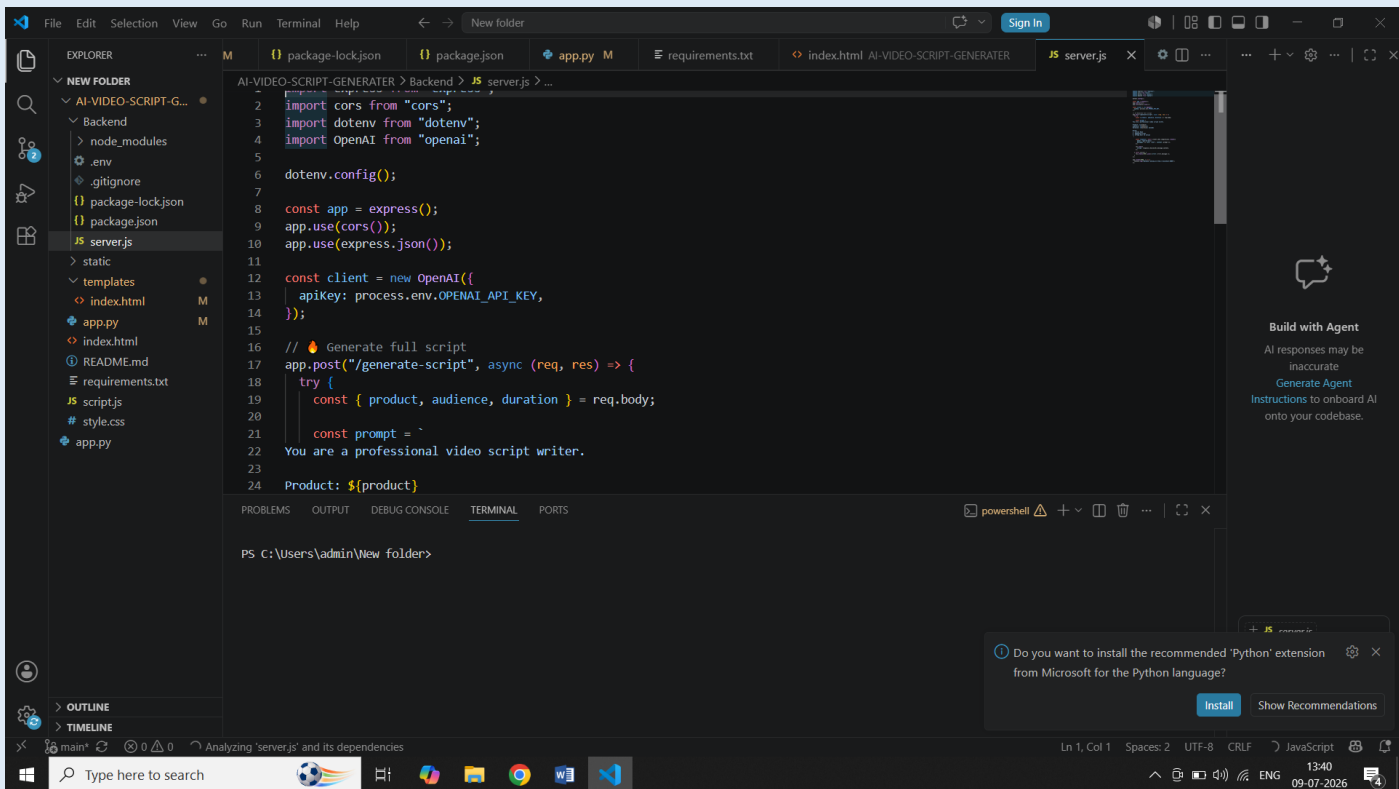
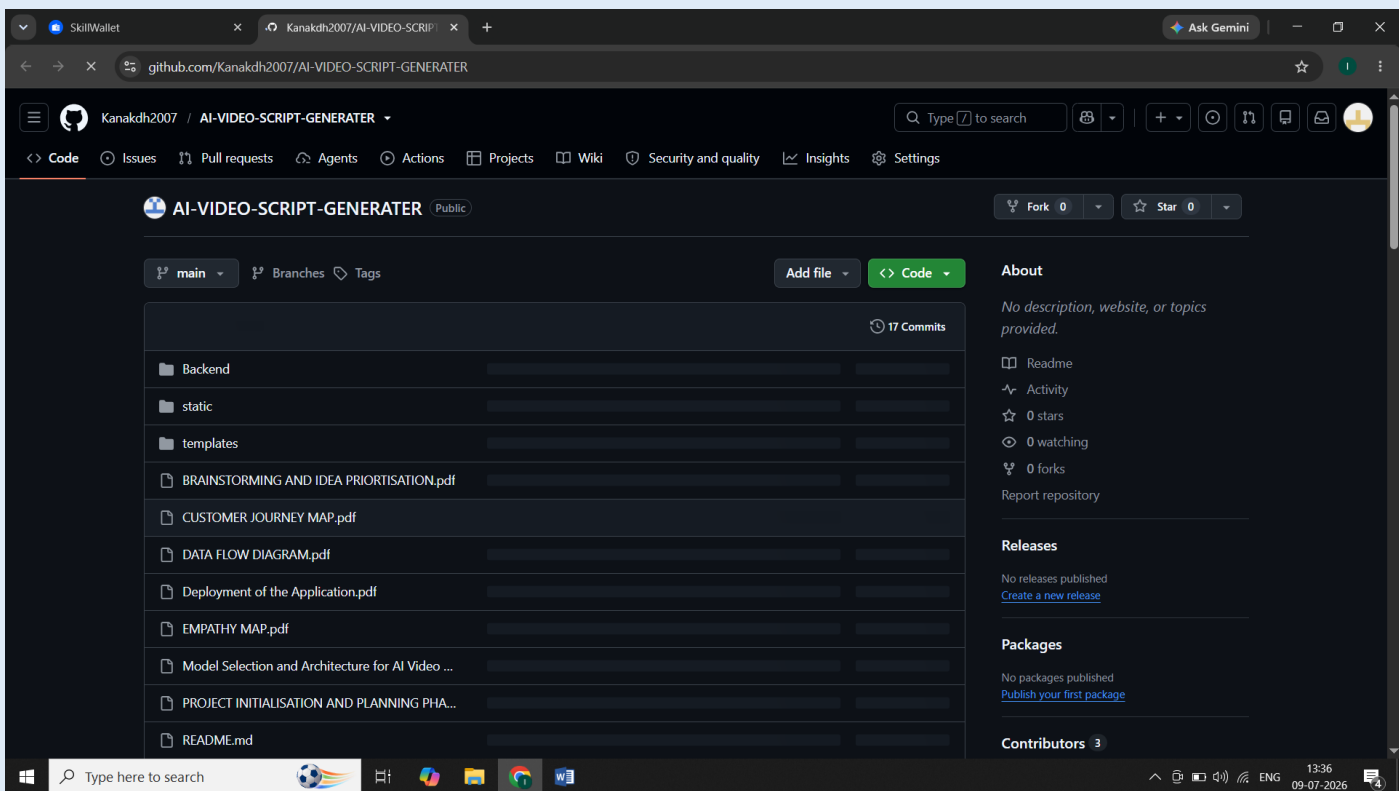
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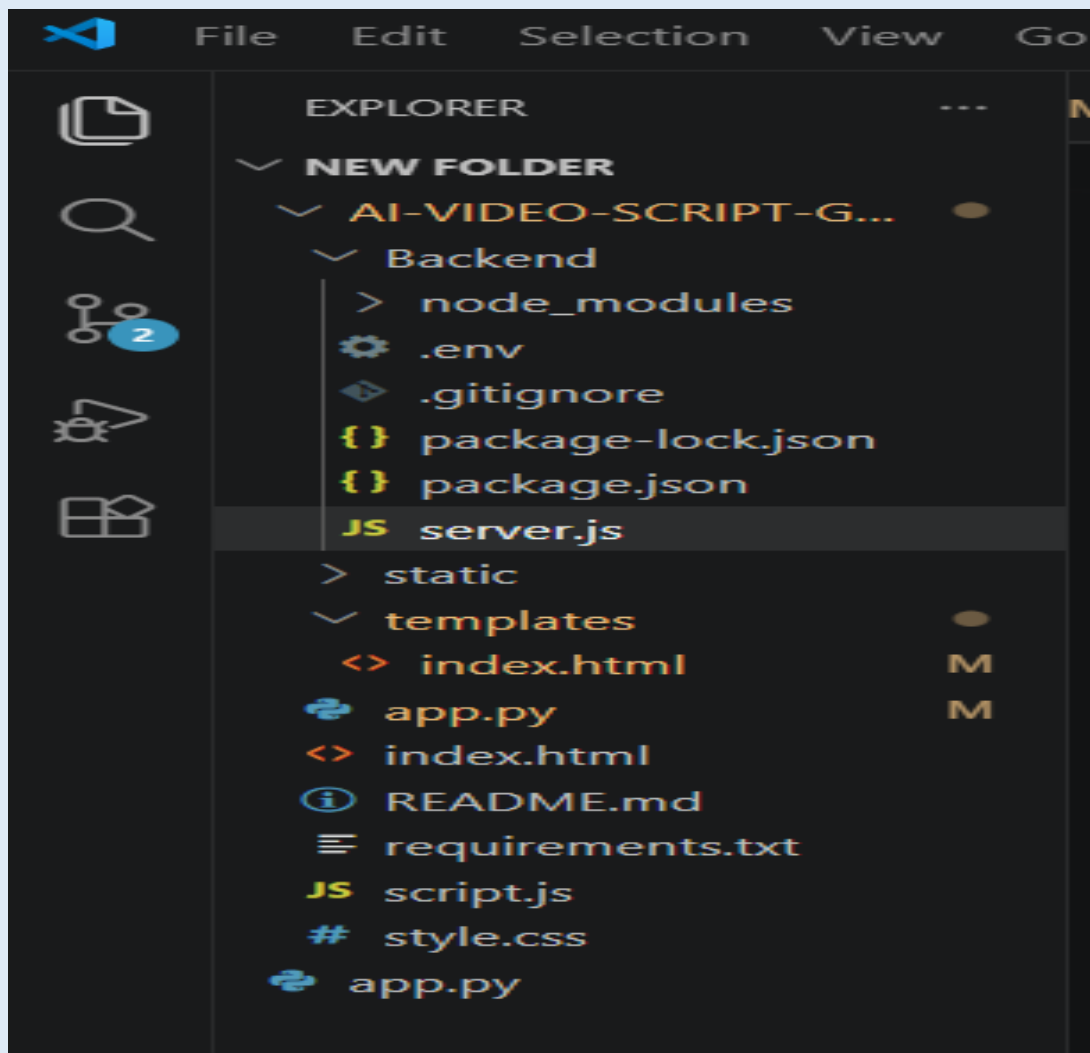
Total Subtasks: 0

Technical Architecture

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TEAM CONTRIBUTION

MEMBER	CONTRIBUTION
ISHA SAINI	Research, AI model selection, project report
BALJEET KAUR	Backend development using Flask, Frontend design
KANAK DHINDHWAL	Task assigning , coordination

References

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 2. Python Official Documentation
 3. Flask Official Documentation
 4. HTML Documentation
 5. CSS Documentation
 6. JavaScript Documentation
 7. GitHub Documentation
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